Measuring shade on a city street

A paired comparison of pavement surface temperature

Methods

We measured surface temperature on two sections of the same street.

One section was shaded by trees. The other was exposed to sunlight.

Temperature difference

We subtracted the shaded reading from the exposed reading.

$$\Delta T = T_{\text{sun}} - T_{\text{shade}} \tag{1}$$

A positive difference indicates a cooler shaded surface.

Interpreting the measurements

Results

At noon the exposed pavement was warmer than the shaded pavement.

Surface temperature describes the pavement itself.

Limits

These measurements do not establish a difference in air temperature.